ST. JOSEPH'S COLLEGE OF ENGINEERING

(An Autonomous Institution) • OMR, Chennai - 119

LAB RECORD

Professional English Lab Integrated

DEPARTMENT OF HUMANITIES & SCIENCES • AFFILIATED TO ANNA UNIVERSITY • CHENNAI

CLASS CODE: EN25C09

STUDENT DETAILS

Student Name: _____

Roll No: _____

Date: _____

Subject Name: _____

Self Introduction: Written Script and One-Page Profile

FATIN AFSA M

Aspiring AI Engineer

Links	GitHub: https://github.com/fatin-afsa-op LinkedIn: https://www.linkedin.com/in/fatin-afsa-982058439/

Summary

I am a first-year Computer Science and Engineering student at St. Joseph's College of Engineering, Chennai. I am interested in programming, artificial intelligence and building useful software. I am currently strengthening my skills in C, Python and core computer science concepts.

Introduction	I enjoy learning programming step by step and solving problems through practice. My current learning projects include a Simple Calculator and a Fake News Detection System in C. I am also exploring DermaTriage, a healthcare-focused hackathon project. My career goal is to become an AI Engineer.
Skills	C, Python, HTML/CSS and AI fundamentals (currently learning)
Education	B.E. Computer Science and Engineering, First Year

Impressive Personality: Biographical Note

SUNDAR PICHAI

Chief Executive Officer of Google and Alphabet

Sundar Pichai, whose full name is Pichai Sundararajan, is an Indian-born technology executive who leads Google and its parent company, Alphabet. He grew up in Chennai, Tamil Nadu, and developed an interest in technology and engineering. He

studied metallurgical engineering at the Indian Institute of Technology Kharagpur. He later earned a master's degree from Stanford University and an MBA from the Wharton School of the University of Pennsylvania.

Pichai joined Google in 2004. During his early years at the company, he helped lead the development of products such as Google Toolbar and Google Chrome. His work in product management and engineering later expanded to include several major Google products and platforms. He became the Chief Executive Officer of Google in 2015 and later became the Chief Executive Officer of Alphabet in 2019.

As a technology leader, Sundar Pichai has overseen the development of products and services used by people around the world. In recent years, Google has placed increasing emphasis on artificial intelligence and other advanced technologies. His career demonstrates the importance of continuous learning, technical knowledge, teamwork and thoughtful leadership.

I admire Sundar Pichai because his journey shows how education, curiosity and steady effort can lead to meaningful contributions in technology. His work inspires me to keep learning and to pursue my goal of becoming an AI Engineer.

Summary of Lecture Notes

NPTEL: Enhancing Soft Skills and Personality (Prof. T. Ravichandran, IIT Kanpur)

Week 1: Course Introduction and Growth Mindset

The course opens with two preview lectures that recap an earlier course, “Developing Soft Skills”. That course dealt with perception, self-management, excellence, conflict, stress, habits, listening, telephone skills, technology, communication barriers, non-verbal communication, presentations, reading and trust. The big new topic here is mindset: a person’s habitual attitudes and ways of reading a situation.

In the shoe-salesman story, nobody wears shoes. Is that “no chance” or a “splendid opportunity”? The same situation draws opposite responses. Drawing on Carol Dweck, the lectures set a fixed mindset against a growth mindset. People with a fixed mindset avoid challenges, resist change, dwell on their limits, reject criticism and stop learning once school is over. Growth-mindset people welcome challenges, learn from failure and treat learning as lifelong.

Learning mindsets form early. A belief like “I’m not smart enough” limits a person, but it isn’t a fact about their ability. The suggested “secrets” are to behave like a baby (open and unafraid), put up with critical people, reject negative stereotypes and stereotype threat, and look for a sense of belonging. The Hawthorne effect is offered as evidence that belonging and attention motivate people more than incentives do.

Week 2: Time Management, Procrastination and Assertiveness

Time can’t really be managed. What we manage is our habits and our perception of it. Monochronic people stick to rigid schedules, while polychronic people stay flexible. Others tend to judge our personality by our punctuality and by how we behave under deadline pressure. To control time, log it, look for “time leakage” (interruptions, indecision, poor organisation, social media, unmanaged email) and make room for productive work. Working at your biological peak time helps. Covey’s urgent/important quadrants and the 80/20 Pareto principle help with prioritising, and “time stealers” such as waiting and travel can be turned into “time gifters.”

The lectures trace procrastination to Parkinson’s law, fear, laziness, perfectionism and addiction, along with the plain feeling of being overwhelmed. The remedies are:

- Eat the elephant bite by bite.
- Do the “frog” (the hardest task) first.
- Use the Kaizen one-minute principle.
- Keep a clean workspace.
- Set goals you actually want and declare them publicly.
- Reward yourself.
- Use the Zeigarnik effect: once you start, you feel compelled to finish.

The week ends with a first look at assertiveness, or not saying “yes” when you mean “no.”

Week 3: Assertiveness, Anger and Positive Thinking

The lectures compare passive, assertive and aggressive people on body language, listening, feelings and outcomes. Assertiveness means stating your rights without violating anyone else's, and it ideally ends in a win-win. Practical steps include checking your relationships, dropping self-debilitating thoughts, avoiding discouraging people, following Stanley and Nancy's commandments and learning four ways to say no.

On anger, Aristotle's point is that the hard part is being angry at the right person, to the right degree, at the right time. Techniques include postponing your reaction, counting, deep breathing, physical exertion, Lincoln's unsent letter, silence and channelling anger into creative work. The lectures then turn to positive thinking through the story of two hospital patients and a "window" that faced a blank wall. Changing negative thinking takes desire and determination, introspection and discarding limitations. It also means building a positive self-image through repetition and visualisation, and bringing body, mind and world into sync.

Week 4: People Skills and Being Attractive

Forty traits make others dislike you, and their opposites make people like you. Examples include egotism, aggression, pessimism, fault-finding, gossiping, false promises, moodiness and favouritism, set against selflessness, optimism, giving credit, politeness, trustworthiness, honesty and humility. For dealing with gossip, Socrates' three-filter test (truth, goodness, utility) is suggested.

Attractiveness has physical, mental, emotional and spiritual layers. Physically, it means neat footwear, a fit body, good diet and sleep, and a signature personality. Mentally, it means intellectual charm and a touch of mystery. Emotionally, it means remembering names, valuing yourself and being yourself. Spiritually, it comes down to mindfulness, deeper connection, helping without expectation and "paying it forward."

Week 5: English Skills and Common Errors

English is the global lingua franca, and avoiding common errors is presented as part of soft skills. Using Sonia Sharma's Common Errors in English, the lectures work through repeated self-scored exercises and explanations. Topics include confused word pairs (resume/résumé, principal/principle, alone/lonely), Indianisms (passed out, joint family, cousin brother) and subject-verb agreement (collective, uncountable and plural-form nouns). The lectures also cover prepositions (since/for, at/in, by/on), articles, collocations and word choices like weather/climate. Twain and Branson are quoted on learning by exposure rather than by memorising rules.

Week 6: Humour and Money

Humour reduces stress, improves health and helps communication. It works at three levels: a smile, hearty laughter, and sarcasm, which is best avoided. Adults laugh far less than children (about 4 times a day versus 300). At work, humour resolves conflict, boosts creativity and productivity, and lets you speak the truth lightly. Suggestions include a joke board, a humour club and a humour hall.

The focus then turns to money and personality. How you treat rich and poor people, how you borrow and lend, and how you handle trust all reveal emotional maturity and integrity. Wealth-building advice includes tracking income and expenses, separating need from greed, avoiding impulse buys, and

becoming indispensable through devoted, excellent work, with Michelangelo, Martin Luther King and Lincoln as examples. Real wealth also includes things money can't buy.

Week 7: Health and Love

Health affects personality, first impressions and employability, since healthy people are seen as disciplined and reliable. Managing it rests on:

- Regular exercise (morning is best, with tips for sticking to it).
- Diet: regular meals, a heavy breakfast and a light early dinner, fruits and vegetables, about 2 litres of water, and no white foods or soft drinks.
- Sleep: 7-8 hours, a good sleep environment, and avoiding gadgets, late heavy meals and alarms.

Love is defined broadly, drawing on Shakespeare, D. H. Lawrence, Gibran, Lovelace and Corinthians: real love is constant, free, patient and unconditional. Managing love means reading body language and timing, letting relationships mature through stages, handling rejection gracefully and keeping romance separate from work.

Week 8: Ethics, Etiquette, Mind, Memory and Environment

Values, ethics and etiquette make up character ("when character is lost, everything is lost"). Professional ethics combine written codes with society's moral expectations. Poor etiquette and business etiquette are covered in detail: introductions, not interrupting, confidentiality, not criticising your company, and praising in public but reprimanding in private.

The mind is powerful but scattered, and it can be trained through auto-suggestion, positive self-talk and effective thinking (gather facts, sleep on it, trust your gut). Memory techniques include association through absurd images, remembering names and faces, number-image links, memory aids and mental exercise. The final lecture urges care for the environment, with practical steps (save water and electricity, avoid plastic, use public transport, plant trees) and a shift from anthropocentric to eco-centric thinking. It closes with Gandhi's line that the earth provides enough for everyone's need but not everyone's greed.

SWOC Presentation

Strategic Career Roadmap: SWOC Analysis (slides)

AI Engineering — My Perspective

My Career Direction

I see myself moving toward AI Engineering because I want to combine software development, artificial intelligence, product thinking, and real-world problem solving. I do not want to learn AI only as theory; I want to understand how an idea can become a useful product that people can actually use.

My Strengths

Curiosity — I naturally ask questions beyond the basic answer. I want to understand why something works, how it works, and how it can be improved.

Builder mindset — I prefer learning by creating. Building websites and experimenting with technology helps me turn concepts into practical skills.

Creativity and uniqueness — I care about making my ideas different instead of simply copying common solutions. This is especially useful in hackathons and product development.

Competitive spirit — Competitions and hackathons motivate me to solve problems creatively, work under constraints, and present ideas effectively.

Product thinking — I am interested not only in code, but also in users, frontend, backend, features, usefulness, and how a complete product works.

Willingness to learn — I am comfortable exploring unfamiliar technologies and asking for explanations when I do not understand something.

My Weaknesses

I can explore too many directions — My curiosity can sometimes make me switch between technologies and ideas before going deep enough into one.

I need stronger technical depth — I am still developing my foundations in programming, AI, backend systems, databases, APIs, deployment, and system design.

Risk of depending too much on AI tools — AI tools can make development faster, but I need to make sure I understand the generated code and can debug and modify it myself.

My portfolio needs deeper projects — The websites I have created are a good beginning, but I need projects that demonstrate stronger engineering, AI integration, deployment, and problem-solving ability.

Consistency is important for me — My progress will depend not only on how intensely I learn, but on whether I continue developing the same core skills over a long period.

My Opportunities

AI + Software — I can combine web development with AI instead of choosing only one field. This can help me build complete AI-powered applications.

Hackathons — Hackathons can help me practice problem identification, rapid prototyping, teamwork, pitching, and building under real constraints.

Growing AI ecosystem — The rapid development of AI APIs, models, agents, cloud platforms, and developer tools creates many opportunities to build new products.

Portfolio building — I can gradually turn my projects into a professional portfolio that demonstrates what I can build rather than only what I have studied.

Long-term product development — I can learn to move from an idea to a user interface, backend, AI system, database, deployment, testing, and continuous improvement.

My Challenges

Rapidly changing AI — AI tools and technologies change very quickly, so I need to focus on fundamental concepts rather than becoming dependent on one particular tool.

Strong competition — Many students will claim to know AI. I need to differentiate myself through real projects, deeper understanding, execution, and product thinking.

Keeping up with technology — AI Engineering requires continuous learning across software, AI, cloud, security, databases, and user experience.

Balancing breadth and depth — I need enough breadth to understand complete products while developing real depth in the technologies that form my core skill set.

My Personal Strategy

My goal is not to become someone who simply says, “I know AI.” I want to become someone who can take a real problem, understand the user, design a solution, build the software, integrate AI where it genuinely adds value, deploy the product, and improve it using feedback.

My Development Path

Strengthen programming and problem-solving fundamentals.
Build stronger web development skills and understand frontend–backend architecture.
Learn APIs, databases, authentication, Git/GitHub, and deployment.
Learn AI and machine-learning fundamentals instead of relying only on AI tools.
Integrate AI into practical applications and understand how AI systems work behind the interface.
Build distinctive portfolio projects and use hackathons to test my skills.
Develop product thinking: identify real problems, validate ideas, design useful experiences, and iterate.

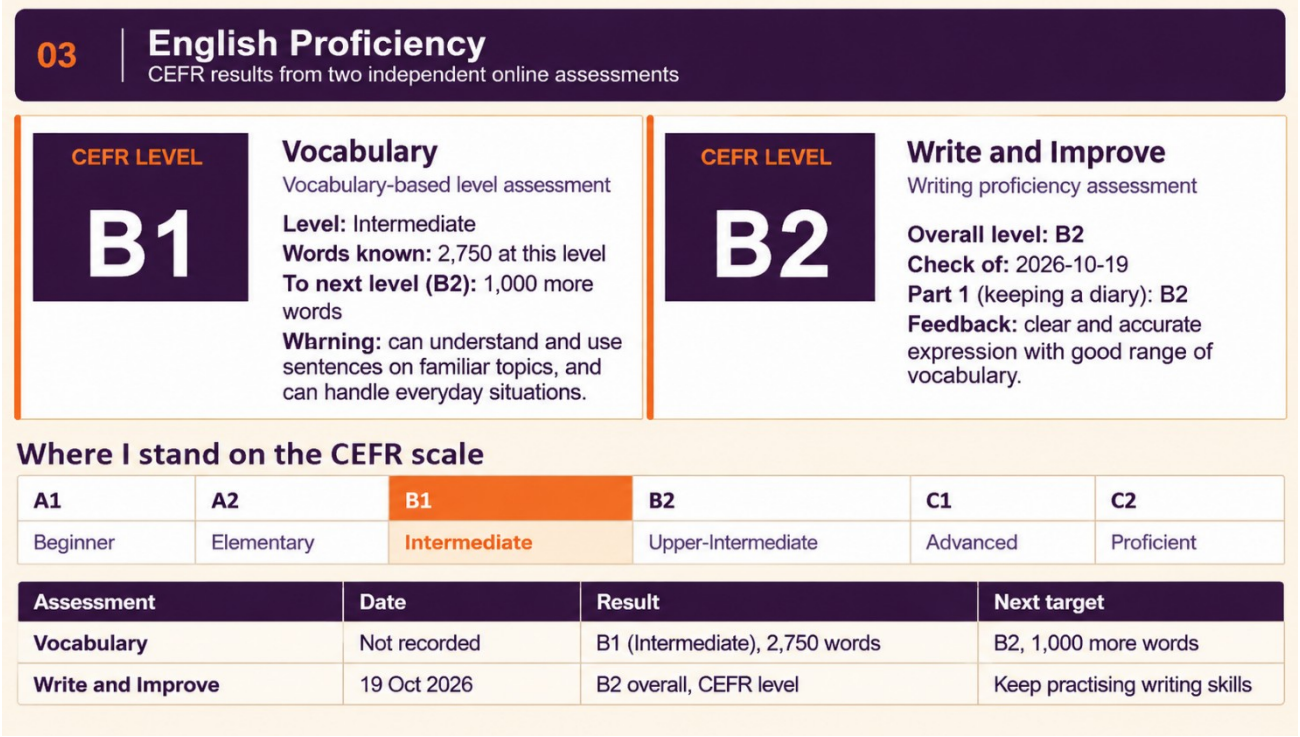
My SWOC in One Statement

My biggest strength and biggest challenge are both my curiosity. I am interested in many technologies and ideas. If I allow that curiosity to scatter my focus, it can slow my progress. If I channel it into a structured learning path and consistently build, it can become one of my strongest advantages.

My target: “Give me a real-world problem. I want to understand the user, design the product, build the software, integrate AI where it actually helps, deploy it, and improve it from real feedback.”

Write and Improve / Speak and Improve

CEFR score card



Professional English Practice Worksheets – Solved

Course Code: EN25C09 | Subject: Professional English – I (Lab Integrated) | Worksheet 1: Model Exam I (Parts A, B, C), followed by Worksheet 2: Model – I Practice Worksheet.

Worksheet 1 (EN25C09 Model Exam I) – Part A (15 × 1 = 15 Marks)

1. **Answer:** monitors
2. **Answer:** recorded
3. **Answer:** stronger
4. **Answer:** must 5. **Answer:** can
6. **Answer:** The laboratory developed a new testing device that measures material strength with high precision.
7. **Answer:** rises
8. **Answer:** would not have failed
9. **Answer:** The mechanics installed dual damper springs to isolate chassis resonance.
10. **Answer:** optimizing

11. **Answer:** 50 mL of hydrochloric acid was poured into the titration flask.
12. **Answer:** smoothly
13. **Answer:** The project supervisor asked the engineer whether he had calibrated the pressure sensor before testing.
14. **Answer:** The incorrect noun form is "equipments", because "equipment" is uncountable. Correct sentence: The laboratory purchased several new pieces of equipment for conducting material testing.
15. **Answer:** worse

Worksheet 1 – Part B (10 × 2 = 20 Marks)

1. **Answer:** The engineer calibrated the pressure sensor and verified its accuracy.
2. **Answer (Portability):** A laptop is more portable than a desktop computer because it is lighter and can be carried anywhere.

   Answer (Screen Size): A desktop computer usually has a larger screen than a laptop, so it is more comfortable for long hours of detailed work.
3. **Answer:** The engineer designed a new cooling system, which reduces the temperature of industrial machines.
4. (a) **Answer:** If the battery voltage drops below the required level, the device will shut down automatically.

   (b) **Answer:** If the technician had checked the connections earlier, the fault would have been detected before the breakdown.
5. **Answer:** The solution was heated to 80°C before testing.
6. **Answer:** The project manager asked the technician whether he had inspected the pressure valves before starting the compressor.
7. (a) **Answer:** The engineer received a lot of feedback on the new design. ("Feedback" is uncountable, so it cannot take "many" or the plural form "feedbacks".)

   (b) **Answer:** The students received several pieces of useful advice from their teacher. ("Advice" is uncountable, so "advices" is wrong.)
8. (a) **Answer:** The technician wears safety gloves to protect his hands.

   (b) **Answer:** The cooling fan operates continuously to prevent the motor from overheating.
9. (a) **Answer:** efficiently

   (b) **Answer:** carefully

10. (a) **Answer:** can

    (b) **Answer:** must not

Worksheet 1 – Part C (5 × 13 = 65 Marks)

Question 26 (a) Safety instructions for safe and responsible use of the internet

1. Use a strong, unique password of at least twelve characters, mixing letters, numbers and symbols, for everyaccount.
2. Enable two-factor authentication on your email, banking and social media accounts.
3. Do not click on links or open attachments from unknown or suspicious senders.
4. Never share personal details such as OTPs, passwords, Aadhaar or bank details with anyone online.
5. Check that a website address begins with "https" and shows a padlock before entering any sensitiveinformation.
6. Keep your operating system, browser and antivirus software updated at all times.
7. Avoid banking or shopping on public Wi-Fi; use a trusted VPN if you must connect.
8. Report cyberbullying, online fraud and phishing attempts immediately to a trusted adult, the platform, or theNational Cyber Crime Helpline (1930).

Question 27 (a) Descriptive essay: A traditional art form in my region and how modern initiatives can preserve it

Selected Question: 27.a Bharatanatyam: The Dance That Carries Tamil Nadu's Soul

Among the many traditional art forms of Tamil Nadu, Bharatanatyam is the one that has carried our culture across the centuries. It began inside the temples, where devadasis performed it as an offering to the deity. The epic Silappathikaram describes classical dance in detail, and the carvings on the Brihadeeswarar Temple at Thanjavur and the Chidambaram Nataraja Temple show its postures in stone. In the nineteenth century, the Thanjavur Quartet organised the repertoire into the form we see today.

Traditionally, a dancer trains for many years under a guru. A performance blends nritta (pure rhythm), nritya (expressive storytelling) and natya (drama), told through hand gestures called mudras, facial expressions and Carnatic music. The dancer wears a bright silk costume, temple jewellery and ankle bells called salangai. The first full solo performance, the arangetram, marks the end of a student's formal training.

Today, many young people feel the art form is old-fashioned and too expensive to learn. Modern initiatives can change this. Schools and colleges can include classical dance in cultural clubs and the curriculum, as institutions like Kalakshetra in Chennai already do. Online classes, recorded performances and social media reels can reach learners in small towns. Festivals such as the Margazhi season in Chennai give young artists a stage, and scholarships can help families who cannot afford training.

Bharatanatyam is more than entertainment; it is a living record of our history, music and devotion. If we support it through education, technology and public platforms, the next generation will carry the dancing bells forward.

Question 28 (b) Formal letter of request to the Principal

Selected Question: 28.b

Fatin Afsa M
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal
St. Joseph's College of Engineering

OMR, Chennai – 600 119

Subject: Request for permission to attend a technical workshop on Artificial Intelligence and Emerging Technologies

Respected Sir/Madam,

I am a first-year B.E. Computer Science and Engineering student of your college. I have come to know that [Name of Institution] is organising a one-day technical workshop on Artificial Intelligence and Emerging Technologies on [date]. I request your kind permission to attend this workshop.

My objective in attending is to learn how artificial intelligence is applied in real engineering problems and to gain hands-on exposure to tools that are not covered in detail in our first-year syllabus. I also hope to meet experts and students from other institutions and learn from their experience.

I expect to learn the basics of machine learning, practical uses of AI in software and cloud systems, and the latest trends in emerging technologies. After the workshop, I will share the notes and key points with my classmates and use what I learn in my Python projects.

I request you to kindly grant me on-duty leave for the day of the workshop and permit me to travel to the venue. I will make up for any classes missed and will submit the participation certificate to the department after the workshop.

I would be grateful if you could grant me this permission.

Thank you for your kind consideration.

Yours obediently,
Fatin Afsa M
(Register No.: 26CS1140)

Question 29 (a) Analytical essay: AI-based assessment versus traditional teacher-based assessment

Selected Question: 29.a AI-Based Assessment versus Traditional Teacher-Based Assessment in Engineering Education

Assessment decides how students learn, so the choice between AI-based and teacher-based methods matters in engineering education. Each has clear strengths and weaknesses.

In terms of accuracy, AI tools are consistent. An automated system can evaluate hundreds of code submissions or objective answers using the same rules every time, without getting tired. Teachers, on the other hand, can understand a student's reasoning in a long derivation or a design report, which software often misses. Therefore, AI is accurate for objective answers, whereas teachers are better for open-ended ones.

Feedback time is where AI wins most clearly. Students receive results and error explanations within seconds, while manual evaluation of a large class may take weeks. Quick feedback lets students correct their mistakes while the topic is still fresh. Conversely, a teacher's comments arrive more slowly but are usually more personal and motivating.

Regarding fairness, AI does not show favouritism or mood-based marking. However, it can carry hidden bias from the data it was trained on, and it may penalise students who give unusual but valid solutions. A teacher can notice such cases and adjust the marks.

Finally, human judgement remains important. Engineering involves creativity, ethics and teamwork, which cannot be reduced to a score.

In conclusion, neither method is perfect alone. The best approach is a blend, where AI handles routine and objective checking quickly, and teachers review complex work and guide students. This combination keeps assessment fast, fair and meaningful.

Question 30 (b) Personal email: Orientation programme

Selected Question: 30.b

To: [friend's email address]

Subject: My first-year orientation programme at college

Dear [Friend's name],

How are you? I hope you are doing well and enjoying your college life too. I have been wanting to write to you about my orientation programme, so here it is.

The programme was held for the first-year engineering students at our college. It began with a welcome address by the Principal, followed by talks from senior faculty and some alumni. We also had fun icebreaker games and team activities, which helped me make new friends within the first day itself.

The best part was the tour of the academic facilities. We visited the library, the computer labs, the workshops and the seminar halls, and the staff explained how to use each of them. We were also told about the college rules on attendance, discipline, dress code and examinations, so now I know exactly what is expected from us.

Overall, the programme made me feel much more comfortable in the new environment. I understood how college life works and how to plan my studies, and I felt ready to start my academic journey with confidence.

Write back soon and tell me about your college too. Take care!

Your friend,
Fatin Afsa M

Worksheet 2 (Model – I Practice Worksheet) – Exercises 1 to 15

Exercise 1: Simple Present Tense (Subject-Verb Concord)

1. recognize
2. requires
3. transmit
4. agrees
5. form

Exercise 2: Simple Past Tense

1. conducted
2. deployed
3. opened
4. published
5. detected

Exercise 3: Comparative Adjectives

1. higher
2. safer
3. faster
4. more durable
5. lower

Exercise 4: Modals of Obligation and Necessity

1. must
2. must
3. have to
4. must

5. has to

Exercise 5: Modals of Capability and Possibility

1. can
2. may
3. can
4. can
5. can

Exercise 6: Relative Clauses

1. The civil engineer who designed the suspension cable bridge received the national infrastructure award.
2. The spectrometer, which analyzes infrared emission spectra, is mounted inside the cleanroom vacuumchamber.
3. The technical specialist whose patent on pneumatic grippers was approved was reassigned to the roboticslab.
4. This microcontroller, which features ultra-low power sleep states, is ideal for battery-operated remotesensors.
5. The field researchers who surveyed the wind turbine farm submitted their acoustic emission assessmentyesterday.

Exercise 7: If-Clauses (Type 1)

1. exceeds
2. fails
3. engages
4. rise
5. registers

Exercise 8: If-Clauses (Type 3)

1. had started
2. had lubricated
3. would have been prevented
4. would have tripped
5. had remained

Exercise 9: Infinitives of Purpose

1. The mechanics installed dual damper springs to isolate chassis resonance.
2. Technicians purge the fuel line with dry nitrogen gas to remove residual flammable vapors.
3. We introduced an optical encoder to determine spindle velocity accurately.
4. The developers wrote an automated load testing script to simulate ten thousand concurrent connections.
5. Gold plating is applied to terminal connectors to prevent atmospheric corrosion.

Exercise 10: Gerunds after Prepositions

1. redesigning
2. replacing
3. reducing

4. cleaning
5. filtering

Exercise 11: Impersonal Passive Voice

1. 50 mL of hydrochloric acid was poured into the titration flask.
2. The robotic arm was tested under a 50 kg dynamic payload.
3. The optical pyrometer was calibrated prior to ignition.
4. Severe cavitation damage was observed along the impeller blades.
5. Volatile solvents are stored in flame-resistant containment cabinets.

Exercise 12: Adverbs of Manner

1. precisely
2. periodically
3. rapidly
4. firmly
5. smoothly

Exercise 13: Reported Speech

1. The quality manager asked the technician whether the insulation resistance test had been documented.
2. The lead architect enquired whether the server cluster could handle double the transaction throughput.
3. The field auditor asked the safety engineer whether he had inspected the gas leakage relief lines.
4. The client asked the team whether the firmware update would fix the intermittent battery drain.
5. The instructor asked the lab group why the multimeter had shown fluctuating voltage readings.

Exercise 14: Countable and Uncountable Nouns

1. Incorrect: "machineries" and "equipments". Correct: The procurement officer purchased several new piecesof machinery and equipment for the workshop.
2. Incorrect: "datas". Correct: The laboratory generated a huge amount of test data during the wind tunnel trials.
3. Incorrect: "softwares". Correct: We need three more software licenses to complete the finite element analysison the model.
4. Incorrect: "informations". Correct: The field report provided a lot of valuable information regarding the soilcomposition.
5. Incorrect: "apparatuses" and "glasswares". Correct: All chemical apparatus and glassware must be washedthoroughly after the experiment.

Exercise 15: Comparative Adjectives (Irregular Forms)

1. worse
2. better
3. farther
4. less
5. further

Worksheet 2 – Part B (Exercises 1 to 10)

Exercise 1: Simple Past Tense Conversion

1. The technician inspected the solar arrays and recorded the peak power conversion value.
2. The chemical analyzer detected trace mercury levels and sent an automated alert to the environmentaldashboard.
3. Maintenance crews lubricated the heavy conveyor bearings and replaced the worn rubber belts before themorning shift.
4. The supervisory software polled field sensors every five milliseconds and archived temperature trends into asecure database.
5. The testing team applied a 500-volt dielectric pulse across the stator winding and monitored for insulationbreakdown.

Exercise 2: Comparative Sentences

1. Cloud storage has a lower initial deployment cost than on-premise physical servers because no hardware hasto be bought. It is also more elastic, since its capacity can be increased within minutes.
2. Fiber-optic cables offer a much higher data transfer bandwidth than copper twisted-pair cables. They are alsoless susceptible to electromagnetic interference than copper cables.
3. Electric vehicles produce lower tailpipe emissions than internal combustion engine vehicles, as they releasenone at all. However, EVs generally have a higher initial purchase price than ICE vehicles.
4. Hydraulic actuators produce a greater raw force output than electromechanical actuators. They also carry ahigher risk of fluid leakage, because electromechanical actuators use no fluid.
5. Relational (SQL) databases have a more rigid schema than document (NoSQL) databases. Documentdatabases usually provide higher horizontal query throughput than relational databases.

Exercise 3: Relative Clauses

1. The algorithm that analyzes microscopic cracks in turbine blades was patented by the aeronautics team.
2. The lead chemical engineer, who noticed an abnormal pressure spike in chamber two, authorized theemergency boiler shutdown.
3. The laboratory houses a 400 kV impulse voltage generator, where high-voltage insulator puncture tests areconducted.
4. We engaged an external cybersecurity consultant whose firmware vulnerability audit exposed three backdoorports.
5. The supervisory system, which utilizes an artificial neural network, predicts transformer insulation life basedon dissolved gas concentrations.

Exercise 4: If-Clauses

1. (a) If the operating temperature exceeds 85°C, the cooling system will switch on automatically.
   (b) If the operator had pressed the emergency shutdown switch immediately, the accident would have beenavoided.
2. (a) If the supply voltage drops below the 200V threshold, the backup power supply will take over.
   (b) If the network administrator had enabled multi-factor authentication, the unauthorized access would havebeen prevented.
3. (a) If toxic carbon monoxide concentrations surpass 50 ppm in the tunnel, the ventilation fans will start and analarm will sound.

(b) If the mechanical team had checked the bearing lubrication schedule, the bearing would not haveoverheated.

4. (a) If the autonomous guided vehicle (AGV) detects an obstacle within 1 meter, it will stop immediately.
   (b) If the thermal sensor had registered the runaway reaction in time, the explosion would have beenprevented.
5. (a) If the software receives an unauthenticated API token, it will reject the request and return an errormessage.
   (b) If the civil inspectors had reviewed the core sample density tests, they would have detected the weakfoundation before construction.

Exercise 5: Impersonal Passive Voice

1. The crude extract was distilled at a constant temperature of 60°C.
2. The cross-section of the weld seam was polished with fine diamond paste.
3. Fifteen cast-iron pump casings were rejected due to internal porosities.
4. The resistance across the bridge circuit was measured using a calibrated digital ohmmeter.
5. The emergency aerodynamic braking flaps were initiated at an altitude of 3,000 meters.

Exercise 6: Reported Speech

1. The safety supervisor asked the intern whether he had verified the grounding connections on the generator.
2. The plant director asked the site engineer whether the emergency effluent treatment plant had receivedclearance.
3. The IT security lead enquired of the developer why he had disabled the firewall rule on the database server.
4. The client asked the project manager whether the bridge structure could withstand Category 4 hurricanewinds.
5. The research guide asked the postgraduate scholar how many sample trials he had conducted undercryogenic conditions.

Exercise 7: Countable vs. Uncountable Nouns

1. (a) We need to install several new pieces of software and machinery in the CAD laboratory.
   (b) The team collected plenty of useful information from the field test.
2. (a) The technician ordered ten new pieces of apparatus and modern electronic equipment for the antennalab.
   (b) High-speed train operations require advanced transport infrastructure and dedicated electrical power.
3. (a) The environmental sensor gathers extensive atmospheric data and pollution indices every minute.
   (b) The construction crew unloaded tons of sand, cement, and steel debris at the worksite.
4. (a) All electrical hardware and wiring must comply with local safety legislation.
   (b) The data scientist discovered several interesting insights after running the machine learning clustering.
5. (a) Laboratory staff must clear away all glassware and acidic waste before leaving the premises.
   (b) The project failed due to a lack of proper funding and technical research.

Exercise 8: Infinitives of Purpose and Gerund Constructions

1. (a) A pressure relief valve is installed to vent excess boiler steam.
   (b) The automated script runs every midnight to archive legacy server logs.
2. (a) Automotive engineers installed an intercooler to reduce the compressed air intake temperature.
   (b) The technicians analyzed the structural stress points by placing piezoelectric strain gauges along thegirder.
3. (a) High-performance heatsinks feature copper heatpipes to conduct heat away from the silicon die.
   (b) The factory cut water consumption by 30% by implementing a closed-loop recycling filtration system.

4. (a) Safety guards are mounted over rotating flywheel belts to prevent accidental worker contact.
   (b) The algorithm optimizes memory bandwidth by pre-fetching sequential array indices into L2 cache.
5. (a) The electronics lab installed an uninterruptible power supply (UPS) to protect servers against suddenvoltage sags.
   (b) The metallurgy group enhanced surface hardness by diffusing nitrogen atoms into the steel casing.

Exercise 9: Adverbs of Manner

1. (a) automatically(b) flawlessly
2. (a) accurately
   (b) instantly
3. (a) continuously
   (b) cautiously
4. (a) consistently
   (b) catastrophically
5. (a) periodically
   (b) smoothly

Exercise 10: Modals

1. (a) can
   (b) must not
2. (a) must
   (b) may
3. (a) must not
   (b) could
4. (a) should
   (b) must
5. (a) can
   (b) must not

Worksheet 2 – Topic 1: Writing Imperative Safety Instructions

Question 1: Chemistry and Materials Testing Laboratory

1. Wear a lab coat, safety goggles and nitrile gloves at all times inside the laboratory.
2. Handle concentrated acids and alkalis only inside the fume hood.
3. Add acid to water slowly; never add water to acid.
4. Do not smell, taste or touch any chemical directly.
5. Label every container clearly, and never return unused chemicals to the stock bottle.
6. Clean all glassware and testing equipment after each experiment to avoid contamination.
7. Keep food, drinks and open flames away from the work area.
8. Report every spill, burn or breakage to the instructor at once, and use the eyewash station or safety shower inan emergency.

Question 2: High-Voltage Electrical Machines and Drives Laboratory

1. Make sure the main supply is switched off before making or changing any connection.
2. Wear insulated rubber gloves and safety shoes, and stand on an insulated rubber mat near live panels.
3. Never touch exposed terminals, bare wires or rotating parts while the machine is running.
4. Get the circuit connections checked by the instructor before switching on the supply.
5. Know the location of the circuit breakers, fuses and emergency stop button, and keep them in workingcondition.
6. Remove rings, watches and metal jewellery, and avoid loose clothing, to prevent arc flash burns andaccidental short circuits.
7. Keep water and wet hands away from electrical equipment, and use only insulated tools.
8. Increase the motor load gradually, and switch off the supply and discharge the capacitors beforedisconnecting the circuit.

Question 3: Advanced Manufacturing and CNC Machining Workshop

1. Wear safety glasses, safety shoes and close-fitting clothes, and tie back long hair.
2. Keep the machine door and guard closed while the spindle is running.
3. Do not touch the spindle, cutter or workpiece until the machine has stopped completely.
4. Clamp the workpiece and tool holder firmly before starting the program.
5. Use a brush or chip hook, and heat-resistant gloves, to remove hot metal chips, and only after stopping themachine.
6. Wipe up coolant spills immediately, and keep the floor dry to prevent slipping.
7. Run the program in simulation mode first, and keep your hand near the emergency stop button during the firstcut.
8. Never leave the machine unattended or operate it without the instructor's permission.

Worksheet 2 – Topic 2: Free Writing – Festivals and Cultural Celebrations

Question 1: A traditional harvest festival and eco-friendly practices

Pongal: The Harvest Festival of Tamil Nadu and How Eco-Friendly Practices Can Protect Its Spirit

Pongal is the most important harvest festival of Tamil Nadu. It is celebrated in mid-January to thank the Sun God, the rain and the cattle for a good harvest. The festival lasts four days: Bhogi, Thai Pongal, Mattu Pongal and Kaanum Pongal.

On Bhogi, families clean their homes and throw away old things. On Thai Pongal, they draw colourful kolams at the entrance and cook fresh rice and milk in a new clay pot in the open yard. When the milk boils over, everyone shouts "Pongalo Pongal!" The pot is decorated with turmeric plants and sugarcane. The festive meal includes sakkarai pongal, venn pongal, sambar, vadai and payasam, served on banana leaves. On Mattu Pongal, farmers decorate and worship their cattle, and Kaanum Pongal is a day for family visits and community games.

Modern eco-friendly practices can preserve the core spirit of the festival. Instead of burning plastic and old tyres on Bhogi, communities can burn only dry leaves or turn them into compost, which reduces smoke and air pollution. Clay pots, banana leaves and areca-leaf plates can replace plastic and thermocol, and kolams can be drawn with natural rice flour. Villages can organise tree-planting drives, rainwater harvesting and cloth-bag campaigns during the festival, and use solar lamps instead of excess electric decorations.

In this way, Pongal can remain joyful and traditional while staying kind to nature, and the next generation will learn that celebration and conservation can go together.

Question 2: A temple car festival and sustainable measures

The Temple Car Festival: Faith, Food and a Cleaner Celebration

In my hometown, as in many towns of Tamil Nadu, the temple car festival, or Ther Thiruvizha, is the biggest event of the year. Thousands of devotees gather in the streets to watch the deity pulled in a huge, richly decorated wooden chariot, called the ther.

The festival begins with flag hoisting and several days of processions. On the car day, devotees of all ages hold thick ropes and pull the heavy chariot together through the four main streets, chanting prayers while drums and nadaswaram fill the air. This shared effort is a strong symbol of unity, because people from every community take part. Women draw kolams on the road, and families offer coconuts, bananas and flowers to the deity. Volunteers give free food such as pongal, puliyodharai, curd rice and sweet payasam as prasadam, served on leaves.

However, such a large crowd also creates problems. Single-use plastic cups, plates and bags litter the streets, and loudspeakers cause noise pollution. Eco-friendly measures can solve these problems. Organisers can ban single-use plastic and provide steel tumblers, banana leaves and paper bags. Dustbins and waste-sorting points can be placed along the route, and volunteers can collect and compost the food waste. Noise can be controlled by limiting loudspeaker volume and using traditional instruments. Barricades, marked entry and exit lanes, volunteers, and first-aid points can manage the crowd safely.

With these steps, the festival can keep its sacred joy and remain clean and peaceful.

Question 3: Regional folk art and handicraft fairs with renewable energy and zero-waste practices

Folk Art and Handicraft Fairs: Heritage with a Green Future

A regional folk art and handicraft fair is a colourful meeting of tradition and community. Walking through the fair grounds, visitors see Karagattam and Oyilattam performances, hear the beat of the parai drum, and watch artisans make Thanjavur dolls, palm-leaf baskets, clay pots and silk sarees. The air smells of fresh filter coffee, murukku, adhirasam and hot sundal.

These fairs are important because they give artisans a market and help young people see the value of their heritage. Many stalls offer live demonstrations, so children can try pottery or weaving themselves. Local delicacies prepared by village cooks add to the festive atmosphere and support small businesses.

However, large fairs use a lot of electricity and produce a lot of waste. Modern environmental strategies can solve this. Solar micro-grids, made of rooftop panels and batteries, can power the lights, sound systems and stalls without diesel generators. Zero-waste composting can turn leftover food and leaves into manure for local farms. Organisers can use reusable steel plates, clay cups and cloth banners, provide separate bins for wet and dry waste, and recycle paper and cardboard.

If we combine the traditional skill of our artisans with clean energy and responsible waste management, these fairs can protect our culture and our planet together.

Worksheet 2 – Topic 3: Free Writing – Food and Technical Travel Plans

Question 1: Technical study tour to a major hydroelectric dam

A Technical Study Tour to the Mettur Hydroelectric Dam

Our department has planned a one-day technical study tour to the Mettur Dam on the Cauvery River in Salem district. The group of forty students and two faculty members will leave the college by a chartered bus at 4:30 a.m. and travel along the national highway, with a short breakfast halt on the way. The journey takes about six hours, and we will return by night. Students will carry ID cards, water bottles and a notebook.

After registering at the gate, an engineer will take us through the powerhouse. The long concrete dam holds back the huge Stanley Reservoir. Inside the powerhouse, we will see the penstocks, the turbines and the generators, and learn how falling water turns turbines to produce electricity. We will also observe the control room, where operators monitor water level, flow and power output on screens, and the spillway gates used to release extra water safely.

For lunch, we will try local dishes such as hot rice with sambar, fish fry from the river and fresh mango juice, which make the trip a cultural experience as well. In the evening, we will walk through the nearby park and enjoy the green landscape and the calm reservoir.

The visit will help us understand renewable energy, water management and the role of civil, mechanical and electrical engineering in one project. We will follow all safety rules, wear helmets where needed, and stay with our group at all times.

Question 2: Technical study tour to an integrated solar and wind hybrid energy farm

A Technical Study Tour to a Solar and Wind Hybrid Energy Farm

We plan a two-day study tour to the Muppandal wind farm region near Kanyakumari, where wind turbines and solar panels work together. Thirty students and two faculty members will take the overnight train from Chennai to Nagercoil, then travel about one hour by a hired van to the farm. We will stay one night in a budget lodge nearby, and return by train on the second night.

On arrival, a site engineer will give a short briefing on how the hybrid plant works. The wind turbines turn on tall towers, and their blades rotate a generator inside the nacelle. We will watch how the strong coastal wind drives them, and how the blades adjust their angle with the wind speed. In the solar section, we will observe rows of photovoltaic panels, inverters and transformers that feed power into the grid. The control room will show how the two sources balance each other: solar power peaks in the afternoon, whereas wind is strong in the evening and night.

For food, we will taste Nanjil Nadu specialities such as fish curry with rice, banana chips, avial and fresh coconut sweets. Eating in small local hotels along the route will also help us understand the region's culture.

The tour will show us the practical side of renewable energy, and we will follow safety rules by wearing helmets, keeping away from the turbine bases and obeying the engineers' instructions.

Question 3: Technical field visit to an automated seawater reverse osmosis desalination plant

A Technical Field Visit to a Seawater Reverse Osmosis Desalination Plant

Our class is planning a half-day field visit to a seawater reverse osmosis desalination plant on the East Coast Road near Chennai, such as the Nemmeli plant. About fifty students and three faculty members will travel in two college buses, leaving at 8:00 a.m. and reaching the plant in about ninety minutes. We will carry ID cards and wear closed shoes for safety.

At the plant, the engineers will first explain the process. We will see the intake pumping station, which draws seawater through long pipes under the sea bed. The water then passes through pre-treatment filters that remove sand and suspended particles. Next, high-pressure pumps push the water through the membrane filtration racks, where the salt is separated and fresh water is produced. We will also see the brine discharge outlet, where the concentrated salt water is carefully diluted and released back into the sea to protect marine life. The control room shows the whole process through automated screens.

After the visit, we will have lunch at a local seaside restaurant. The coastal menu includes fish fry, prawn masala, crab curry with steamed rice and filter coffee, which gives us a taste of the Chennai coastal culture.

The visit will help us see how engineering solves water scarcity, how energy is used in the process, and why environmental care in brine disposal is important.

Worksheet 2 – Topic 4: Formal Letters of Request

Question 1: Letter to the Head of the Computer Science Department (lab access for a hackathon)

Fatin Afsa M
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119

1 November 2026

To
The Head of the Department
Department of Computer Science and Engineering
St. Joseph's College of Engineering OMR,
Chennai – 600 119

Subject: Request for permission and lab access after college hours to prepare for a National AI Hackathon

Respected Sir/Madam,

I am writing on behalf of my project team of four first-year students. We have registered for a National AI Hackathon to be held on [date], and we respectfully request permission to use the computer laboratory after college hours.

The objective of our project is to build a simple AI-based solution that solves a real campus problem, using Python and machine learning. We need about three weeks to design, train and test the model, and our home systems are not powerful enough for this.

We require access to the lab between 4:30 p.m. and 7:30 p.m. on weekdays, with Python, Jupyter Notebook, TensorFlow or scikit-learn, Git and a stable internet connection. We promise to follow all lab rules, maintain the equipment carefully, and leave the lab clean and switched off after use. We will also work under the supervision of a faculty member if required.

We would be very grateful if you could grant us this permission.

Thank you for your kind consideration.

Yours obediently,
Fatin Afsa M
(On behalf of the project team)

Question 2: Letter to the Plant Manager (industrial plant visit)

Fatin Afsa M
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Plant Manager
[Name of Factory], Industrial Robotics and Automation Division
[Address]

Subject: Request for permission to conduct an industrial visit for 50 first-year engineering students Respected

Sir/Madam,

I am a first-year engineering student of St. Joseph's College of Engineering, Chennai. On behalf of my class, I request your permission to visit your robotics and automation factory with fifty students and two faculty members on a convenient date in [month].

The intended learning outcomes of the visit are to understand how robots and automated systems work on a real production line, to see how sensors, controllers and robotic arms are combined, and to learn about industrial safety practices and quality control. The visit will also help us connect our classroom learning with practical engineering and inspire us to think about careers in automation.

We will arrive in a group, carry our ID cards, follow all safety rules, and obey your staff's instructions throughout. We would be glad if a short briefing and a guided tour of about two hours could be arranged.

I would be grateful if you could kindly grant us this opportunity. I look forward to your reply.

Yours faithfully,
Fatin Afsa M

Question 3: Letter to the Head of the Mechanical Engineering Department (machine shop access for an electric go-kart)

Fatin Afsa M
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Head of the Department
Department of Mechanical Engineering
St. Joseph's College of Engineering OMR,
Chennai – 600 119

Subject: Request for permission and machine shop access after college hours to build an electric go-kart

Respected Sir/Madam,

Our student team has been selected to take part in an inter-collegiate design challenge to be held on [date], where we must build and test an energy-efficient electric go-kart. We therefore request permission to use the machine shop after regular college hours.

We need the workshop to cut and weld the chassis, fit the motor, battery and braking system, and test the kart on the campus ground. We estimate that we will need about four weeks, working from 4:30 p.m. to 7:30 p.m. on weekdays and during part of the weekends.

We will work only under the supervision of the workshop instructor, wear safety glasses, gloves and shoes at all times, and follow all machine safety rules. We will take care of the tools and keep the shop clean after use.

We would be thankful if you could grant us this permission and allow us to bring honour to the college.

Thank you for your kind support.

Yours obediently,
Fatin Afsa M
(On behalf of the team)

Question 4: Letter to the Principal / Dean of Student Affairs (sponsorship and duty leave)

Fatin Afsa M
I Year, B.E. Computer Science and Engineering
St. Joseph's College of Engineering
OMR, Chennai – 600 119
1 November 2026

To
The Principal / Dean of Student Affairs
St. Joseph's College of Engineering OMR,
Chennai – 600 119

Subject: Request for financial sponsorship and duty leave to present a drone surveillance prototype at the National Defense Innovation Expo

Respected Sir/Madam,

I am happy to inform you that a five-member multidisciplinary team of our college has been selected to present a drone surveillance prototype at the National Defense Innovation Expo, to be held on [dates] at [venue]. We respectfully request financial sponsorship and duty leave for this event.

Our prototype uses a camera-equipped drone with image processing to monitor sensitive areas in real time. Presenting it at a national level will give us valuable feedback from experts and will bring recognition to our college.

We request sponsorship for travel, accommodation, registration and prototype transport, which we estimate at ₹ [amount], and duty leave for [number] days so that we do not lose attendance. We will make up for the missed classes and submit a report after the event.

I would be grateful for your kind support and approval.

Thank you.

Yours obediently,
Fatin Afsa M
(On behalf of the team)

Worksheet 2 – Topic 5: Compare and Contrast Descriptive Essays

Question 1: Electric Vehicles versus Internal Combustion Engine Vehicles

Electric Vehicles versus Conventional Internal Combustion Engine Vehicles

Electric vehicles (EVs) and internal combustion engine (ICE) vehicles are two different answers to the same need, which is transport. Each has clear positives and negatives.

Consider battery life first. An EV battery usually lasts eight to ten years, but replacing it is very expensive. In contrast, an ICE vehicle has no battery problem, though its engine needs regular servicing. As for the manufacturing footprint, EVs create more emissions during production because of battery mining and processing, whereas ICE vehicles have a lighter manufacturing footprint. However, over its lifetime an EV produces far lower emissions, especially when the electricity comes from renewable sources.

Operating costs also differ. EVs have fewer moving parts, so they need less maintenance, and electricity per kilometre is cheaper than petrol or diesel. On the other hand, ICE vehicles cost less to buy, and their repair shops are available everywhere. Fuel prices, however, keep rising and make them costly to run.

Charging network feasibility is the biggest challenge for EVs. Charging stations are still limited, especially on highways and in small towns, and charging takes much longer than refuelling. ICE vehicles, conversely, can be refuelled in minutes at petrol stations spread across the country.

In conclusion, EVs are cleaner and cheaper to run, whereas ICE vehicles are cheaper to buy and more convenient at present. As battery prices fall and charging networks grow, EVs are likely to become the better choice for the future.

Question 2: Remote/Virtual Internships versus On-Site Internships

Remote Internships versus On-Site Internships for First-Year Engineering Students

Internships help early-stage engineering students connect classroom theory with real work. Today, students can choose between remote (virtual) and on-site (physical) internships, and each has advantages and disadvantages.

Hands-on skill acquisition is the biggest difference. On-site internships give direct access to machines, laboratories and real equipment, so students learn practical skills such as wiring, fabrication and testing. Remote internships, in contrast, mostly involve software, simulations and online tasks, which are useful for coding and design but cannot replace physical practice.

In terms of schedule flexibility, remote internships win clearly. Students can work from home, save travel time and cost, and manage studies alongside the internship. On-site internships, on the other hand, demand fixed hours and daily travel, which can be tiring during the semester.

Mentorship is stronger on-site, because supervisors can guide students directly and correct mistakes immediately. Online mentors, however, are often busy and communicate only through scheduled calls or messages, so doubts may stay unresolved for days.

Professional networking also favours on-site internships, as students meet colleagues, build relationships and learn workplace culture naturally. Remote interns may never meet their team face to face, although online platforms still allow them to connect with professionals from different cities.

In conclusion, remote internships are better for flexibility and software skills, whereas on-site internships are better for practical learning, mentorship and networking. A first-year student should ideally begin with a remote internship and move to an on-site one later.

Question 3: Cloud Computing Infrastructure versus On-Premise Data Centers

Cloud Computing versus On-Premise Physical Data Centers for Engineering Enterprises

Engineering enterprises must decide whether to run their systems on the cloud or in their own physical data centers. Both choices have strengths and weaknesses.

Upfront capital expenditure is the clearest difference. Cloud services need almost no initial investment, because companies simply pay for what they use. On-premise data centers, in contrast, require heavy spending on servers, cooling systems, power backup and buildings before any work begins.

As for recurring overheads, cloud bills can grow quickly if usage is not monitored, whereas an on-premise setup has predictable costs once it is built, although it still needs staff, electricity and maintenance. Therefore, cloud suits changing workloads, while on-premise suits stable and long-term demand.

Workload scalability strongly favours the cloud. Companies can add computing power in minutes during simulations or product launches and reduce it afterwards. On the other hand, on-premise systems have fixed capacity, so expanding them takes weeks or months and leaves hardware idle during quiet periods.

Data security and compliance are more complex. On-premise data centers give the company full physical control, which is important for defence, healthcare or confidential design data. Cloud providers offer strong security tools and certifications, but the company depends on a third party and must follow data-location laws carefully.

In conclusion, the cloud is better for flexibility, speed and low starting cost, while on-premise infrastructure is better for control and predictable long-term expenses. Many enterprises now choose a hybrid model that combines the advantages of both.

Question 4: 3D Printing versus Traditional CNC Milling

3D Printing (Additive Manufacturing) versus CNC Milling (Subtractive Manufacturing) in Prototype Fabrication

In prototype fabrication, 3D printing and CNC milling follow opposite principles. A 3D printer builds a part layer by layer, whereas a CNC machine cuts material away from a solid block. Each method has positives and negatives.

Material wastage is the first point of contrast. 3D printing uses only the material needed for the part, so wastage is low. CNC milling, on the other hand, removes a large amount of material as chips, which increases the cost of expensive metals.

When it comes to production speed, the picture changes with batch size. For single prototypes and small batches, 3D printing is faster because it needs no tooling or setup. For larger batch runs, however, CNC milling is quicker, as it can produce identical parts continuously at high speed.

Surface finish and accuracy favour CNC milling. It produces smooth surfaces and very tight tolerances, so parts often need no extra work. In contrast, 3D printed parts often show layer lines and need sanding or polishing, and their accuracy is lower.

Geometric complexity is the strength of 3D printing. It can easily create hollow parts, internal channels and lattice structures that are impossible to cut with a rotating tool. CNC milling is limited by tool access and the direction of cutting.

In conclusion, 3D printing is ideal for complex, low-volume prototypes with little waste, whereas CNC milling is better for accurate, smooth and larger-batch parts. Engineers often use both methods together to get the best result.

Worksheet 2 – Topic 6: Personal Email Writing

Question 1: Email to a high school mentor/teacher about the first semester

To: [teacher's email address]

Subject: My first semester at engineering college

Dear Sir/Madam,

I hope you are doing well. I wanted to write to you because your guidance in school gave me the confidence to start my engineering journey, and I would like to share how my first semester has been.

The biggest change was the move to independent campus life. Nobody reminds us about deadlines now, so I had to learn to manage my own time, plan my study hours and take responsibility for my work. It was difficult at first, but I am getting used to it.

In our laboratory sessions, we worked on our first small projects, such as basic programs in Python and C and simple circuits. Seeing my own code run correctly gave me a great sense of satisfaction, and I realised how different practical learning is from reading theory.

For the end-semester examinations, I have made a timetable that gives equal time to every subject. I revise the notes every weekend, solve previous question papers and practise the lab experiments so that I do not forget the procedures.

Thank you for everything you taught me. I will be very happy to hear from you and to receive any advice for the coming semesters.

With respect and regards,
Fatin Afsa M

Question 2: Email to a high school science teacher about the first term

To: [science teacher's email address]

Subject: Reflections on my first term in engineering

Dear Sir/Madam,

I hope you are keeping well. It has been a few months since I joined engineering college, and I wanted to share my first-term experience with you, since you were the one who made science exciting for me.

The first thing I noticed was how different the hands-on lab coursework is. In school, we mostly saw experiments in a few periods, but here we handle components, tools and software ourselves. I made mistakes in the beginning, but each mistake taught me something and I am becoming more careful and confident.

I also took part in my first technical club activity and a small hackathon. Working with a team to build a solution within a limited time was challenging and thrilling, and I learned a lot about teamwork and debugging from my seniors.

For the upcoming university theory and practical exams, my strategy is simple. I will finish revising each unit at least two weeks before the exams, practise numerical problems daily, and repeat the lab experiments so that I can answer viva questions without fear.

I would be grateful for any advice you may have. Please take care.

Yours sincerely,
Fatin Afsa M

Question 3: Email to an elder sibling/alumnus seeking advice

To: [sibling's or alumnus's email address]

Subject: Need your advice on balancing studies and technical clubs

Dear [Name],

How are you? I hope college and work are going well. I am writing because I need your advice on something I am struggling with.

My day starts at 6:30 a.m., and I attend classes until 4:30 p.m. After that, I go to my technical club meetings for about two hours, and in the evening I try to study and finish my assignments. By the time I am free, I am usually very tired.

My biggest challenge is managing the lab record submissions. Because the records must be completed every week, they pile up, and I often finish them at the last minute, which affects the quality.

Since you have handled the same situation, could you please tell me how you balanced your academics with the clubs? I would also like your tips for semester exam revision, such as how to plan the timetable, which subjects to start with, and how to use previous question papers.

I am looking forward to your reply. Thank you in advance, and please take care.

Your loving brother/sister,
Fatin Afsa M

Worksheet 2 – Topic 7: Travel Planning and Budgetary Itineraries

Question 1: 4-day tour to an aerospace and satellite launch station

A 4-Day Student Technical Tour to an Aerospace and Satellite Launch Station

Our department plans a four-day tour to a satellite launch station for forty students and two faculty members. We will travel by a chartered train coach and local bus, so that the cost is reasonable.

Budget (per student): train fare ₹1,200, local bus ₹400, accommodation for three nights ₹1,500, food ₹1,200, entry and miscellaneous charges ₹400. The total is ₹4,700, which gives about ₹1,88,000 for the full group of forty.

Day 1: Overnight travel and arrival, check-in at the guest house and an orientation briefing. **Day 2:** Visit the launch complex viewing gallery, the rocket assembly area and the mission control centre, where scientists explain how a launch is planned and tracked. **Day 3:** Visit the satellite integration and testing laboratory and the space museum, followed by an interaction with engineers on careers in aerospace. **Day 4:** Morning group discussion, preparation of our tour report, and the return journey.

Safety protocols: Carry your ID cards and keep them visible at all times. Follow the instructions of the security staff, avoid photography in restricted areas, and stay with the group. Wear closed shoes and drink plenty of water. Keep the emergency contact numbers of the faculty coordinators and follow the launch-site rules strictly.

This tour will give us a first-hand view of how rockets and satellites are built and operated.

Question 2: 4-day tour to a major commercial port and maritime dry-dock facility

A 4-Day Industrial Exposure Tour to a Major Commercial Port and Dry-Dock Facility

Our class of forty first-year students and two faculty members will visit a major commercial port and a dry-dock facility over four days. We will travel in a hired bus, and stay in a budget hotel near the port.

Estimated cost per student: travel ₹1,500, stay for three nights ₹1,800, food ₹1,200, entry passes and miscellaneous ₹500. The total is ₹5,000 per student.

Day 1: Travel to the port city, check in and a short orientation. **Day 2:** Technical briefing at the port authority, followed by a visit to the container terminal, where cranes, conveyors and tracking systems handle cargo. **Day 3:** Visit the dry-dock, where ships are lifted and repaired, and see how hull cleaning, welding and engine maintenance are done. An engineer will also brief us on ship stability. **Day 4:** Visit the control tower and the vessel traffic system, a feedback session and the return journey.

Dockyard safety protocols: Wear helmets, safety shoes and high-visibility jackets at all times. Do not go near moving cranes, cargo or the edge of the dock. Stay in the group, obey the guides and never touch machinery. Keep emergency contacts ready, and carry first-aid supplies.

This tour will help us understand how ports and shipyards work in real life.

Question 3: 4-day tour to an electric locomotive manufacturing works and railway signalling center

A 4-Day Technical Study Tour to an Electric Locomotive Works and a Railway Signalling Center

We plan a four-day study tour for thirty-five students and two faculty members to an electric locomotive manufacturing works and an automated railway signalling center. We will book sleeper-class seats on trains for both journeys, and use a local bus for the visits.

Category-wise expenditure per student: rail tickets ₹1,400, local transport ₹300, accommodation for three nights ₹1,500, food ₹1,200, and entry and miscellaneous ₹300. The total is ₹4,700.

Day 1: Overnight train journey and arrival. **Day 2:** Visit the locomotive works, where we will see the shop-floor stages such as shell fabrication, wiring, traction motor assembly and final testing of the engine. **Day 3:** Visit the automated signalling center, where engineers explain how sensors, track circuits and computers control train movement and keep the trains safe. **Day 4:** Interaction with engineers, preparation of a tour report and the return journey.

Site safety rules: Wear helmets, safety shoes and closed clothing in the shop floor areas. Never walk on the railway tracks, and keep a safe distance from cranes and live electrical equipment. Follow the guides' directions and stay with the group. Do not touch any machine or signal equipment, and report any problem immediately to the faculty coordinator.

This tour will connect our classroom learning with real railway engineering.

Project Proposal: Ideation



01 Introduction

LifeLink is a personal safety mobile application that combines journey planning, emergency access, vehicle details, and time-based safety check-ins in one place. It helps people prepare before travelling and respond faster when a journey becomes worrying or an emergency occurs.

Problem addressed

During a stressful situation, users may lose time switching between maps, contacts, alarms, and messaging tools. LifeLink proposes one simple workflow: plan a route, set an expected arrival time, keep emergency options accessible, and confirm safe arrival.

Main objective

To create a simple, privacy-conscious safety prototype connecting route navigation, an SOS alarm, emergency numbers, vehicle trip details, and a journey timer.

PLAN	CHECK IN	RESPOND
Choose a destination and route.	Set arrival time and trusted contacts.	Access SOS and emergency shortcuts.

02 Proposed Solution and Features

The application is organized around one active journey at a time.

1. Mapping and Fixed Routes

Show current location and a route to a destination such as home, college, or a bus stop. Saved routes can be reopened quickly.

2. Emergency Alarm and SOS

A prominent SOS control can activate a loud phone alarm, show emergency-call options, and send an alert with available location data to trusted contacts.

3. Emergency Numbers

Keep police, ambulance, fire services, and trusted contacts easy to reach. For an India-focused prototype, include 112.

4. Vehicle Number and Trip Details

Let users enter a vehicle registration number and basic trip details, then share them with a trusted contact if required.

5. Time Limits and Safety Check-ins

Set an expected arrival time. If the timer expires, the app prompts the user to confirm safety and can send a configured alert.

6. Minimal Data Storage

Avoid continuous location-history storage by default. Keep trip details only as long as needed for the active session and delete temporary data under clear rules.

Important: A phone alarm does not automatically notify police. Remote alerts depend on permissions, connectivity, device settings, and implemented services. Entering a vehicle registration number does not verify the owner's identity.

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03 Working Process

A simple flow makes the concept easy to explain and demonstrate.

STEP 1 — START JOURNEY
Choose destination, route, and optional vehicle details.

STEP 2 — SET SAFETY TIMER
Set expected arrival time and select trusted contacts.

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STEP 3 — FOLLOW ROUTE
Keep map and SOS controls accessible during travel.

STEP 4 — CONFIRM ARRIVAL
Tap "I'm Safe" when you reach your destination.

Emergency response

WHEN SOS IS PRESSED
The app can activate the configured alarm, show emergency calling options, and send an alert with available location information to trusted contacts.

WHEN THE TIMER EXPIRES
The app prompts the user to confirm safety. If there is no response, it can send a preconfigured check-in alert to the selected contact.

Example scenario

A student travelling home selects a saved route, enters the cab registration number, and sets a 30-minute check-in timer. On arrival, the student taps "I'm Safe." If the timer expires first, the app requests confirmation and can notify the selected contact if there is no response.

PLAN YOUR JOURNEY • STAY CHECKED IN • GET HELP FASTER

04 Technology Stack and Feasibility

A first version can be built in stages using widely available tools.

TECHNOLOGY	PURPOSE	REASON FOR CHOICE
Flutter	Mobile app interface	One codebase can support Android and iOS.
Google Maps Platform or OpenStreetMap	Maps and route planning	Provides map and route options for the prototype.
GPS / device location services	Current location	Uses phone location with user permission.
Firebase Authentication	Optional user sign-in	Can simplify account setup for a demo.
Firebase Cloud Messaging	Notifications to registered devices	Useful for alerts; setup and testing are required.
Local notifications and timers	Countdown and check-in reminders	Supports the time-limit feature.
Local database	Temporary trip details	Stores only information needed for the active session.

Proposed development timeline

WEEK 1	WEEK 2	WEEK 3	WEEKS 4–5
Design screens and create map/route flow.	Add SOS screen, emergency shortcuts, and timer.	Add trusted-contact alerts and arrival check-in.	Test permissions, false alarms, connectivity, and deletion.

Feasibility: A first-year CSE team can demonstrate the core workflow in approximately 3–5 weeks, depending on experience and available devices. Reliable background tracking and remote alert delivery need real-device testing and careful permission handling.

05 Novelty, Challenges and Conclusion

The central idea is the connected safety journey—not a claim that every feature is entirely new.

What makes LifeLink different

- **Journey-centred safety**
  Route, timer, arrival confirmation, and SOS are linked in one flow.
- **Missed check-in response**
  A time limit prompts the user to confirm safety and can trigger a configured alert.
- **Vehicle-aware trip details**
  Vehicle registration details can be included in a trip summary shared with a trusted contact.
- **Privacy-conscious design**
  Location history is minimized by default and temporary trip data can be deleted after the session.

Challenges and practical responses

CHALLENGE	PRACTICAL RESPONSE
Weak internet connection	Keep local alarm and emergency-number shortcuts available; show when remote alerts cannot be sent.
False alarms or missed timers	Use a confirmation prompt, clear cancel option, and user-controlled alert settings.
Privacy and misuse	Use consent, limited permissions, secure handling, and automatic deletion of temporary data.
Background app limitations	Test on real phones and explain battery, operating-system, and notification limitations.

Conclusion

LifeLink combines fixed-route navigation, emergency alarms, quick access to emergency numbers, vehicle details, and time-limited safety check-ins. Users select a destination, set an expected arrival time, and choose trusted contacts. If they do not confirm safe arrival, the app can prompt them and send a configured alert. By focusing on timely check-ins and minimal data storage, LifeLink aims to make everyday journeys safer without encouraging unnecessary surveillance.

LIFELINK

ONE TAP. ONE ROUTE. ONE LIFELINE.

LinkedIn and GitHub Profiles

Part A: LinkedIn profile

ABOUT ME: **Hello ,**

I'm Fatin Afsa M, a first-year B.Tech Computer Science and Engineering student at St. Joseph's College of Engineering.

I'm passionate about technology, Artificial Intelligence, software development, and innovation. My ambition is to become an AI Engineer and build intelligent products that solve real-world problems.

I am building my programming foundation through academic and personal projects, including a Simple Calculator and a beginner-level Fake News Detection System in C. I am also working on the DermaTriage hackathon concept.

💻 Technical Skills & Interests

- Python Programming
- C Programming
- HTML, CSS & JavaScript
- Bootstrap
- Web Development
- Artificial Intelligence
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🌱 Professional Strengths

- Communication
- Problem Solving
- Creativity
- Adaptability
- Continuous Learning

🎯 Career Goal

To become an AI Engineer, combining software engineering, artificial intelligence, and product thinking to develop impactful technology solutions.

I believe in learning by building, experimenting, and continuously improving.

Always excited to connect with developers, mentors, students, and technology enthusiasts!

#ArtificialIntelligence #AIProductEngineer #CSE #Python #WebDevelopment #WomenInTech

Featured projects

AI Interview Prep Platform | Preppy Case Study

HTML · CSS · JavaScript · Bootstrap

Biodiversity Assessment Automatizzato | Piattaforma Superiore al Consulente Ambientale - XNatura

HTML · CSS · JavaScript · Python

Security Dashboard designs, themes, templates and downloadable graphic elements on Dribbble

HTML · CSS · JavaScript · Bootstrap · Python

These are illustrative project descriptions, not claims of verified security or AI functionality.

Skills

Category	Skills
Programming	Python, C
Web development	HTML, CSS, JavaScript
Frameworks	Bootstrap
Interests	Artificial Intelligence, Product Development
Soft skills	Communication, Teamwork
Strengths	Problem Solving, Adaptability

Career roadmap

Year	Focus	Target project
2026–27	Programming, web development, Python	Personal portfolio
2027–28	Data structures, AI, ML fundamentals	AI interview assistant
2028–29	APIs, AI integration, databases	AI-powered web product
2029–30	Product engineering, deployment	Complete AI product

GitHub profile README

Hi there, I'm Fatin Afsa M 👋

🎓 B.Tech Computer Science & Engineering Student (2026–2030)

🤖 Aspiring AI Engineer

💻 Beginner Programmer | C and Python Learner

I'm passionate about technology, artificial intelligence, and building digital solutions that solve real-world problems.

👩‍💻 About Me

- 🎓 First-year CSE student at St. Joseph's College of Engineering.
- 🚀 Interested in AI Engineering and software development.
- Learning through C programming projects and a healthcare-focused hackathon concept.
- 🌱 Continuously learning and exploring new technologies.
- 🎯 Working towards becoming an AI Engineer.

🛠️ Tech Stack

Languages: C and Python (beginner level)

Web technologies: HTML and CSS (learning)

Areas of Interest: Artificial Intelligence, Web Development, Product Engineering

🚀 Featured Projects

1. Interview Preparation Platform

A web platform designed to help people from different backgrounds prepare for interviews.

Technologies: HTML, CSS, JavaScript, Bootstrap

2. Landscape Identification & Records Management

A website focused on landscape identification and maintaining organized records.

Technologies: HTML, CSS, JavaScript, Python

3. Secure Banking Workflow Website

A sample banking web application focused on banking operations and protecting sensitive information.

Technologies: HTML, CSS, JavaScript, Bootstrap, Python

🎯 My Career Goals

- Strengthen my programming and software engineering fundamentals.
- Explore Artificial Intelligence and Machine Learning.
- Strengthen programming skills and gradually build practical AI-powered applications.
- Develop strong product development skills.
- Become an AI Engineer.

🌐 Connect With Me

- LinkedIn: https://www.linkedin.com/in/fatin-afsa-982058439/
- GitHub: https://github.com/fatin-afsa-op

⭐ Always learning. Always building. Always improving.